smartwilds-drones — dataset card (review draft)

Rendered from release/hf_smartwilds_20261022/smartwilds-drones/README.md

Dataset Card for SmartWilds Drones



Przewalski's horse
Equus przewalskii

Sichuan takin
Budorcas taxicolor

Milu deer
Elaphurus davidianus

SAM3 detections on drone frames of Przewalski's horse, Sichuan takin, and Milu (Père David's) deer.

Every drone video flown at The Wilds in 2025, with each platform's raw telemetry, and frame-level animal detections placed on the ground where the telemetry allows it.

This is the drone dataset of the **SmartWilds** collection. Each modality has its own dataset and DOI. They share one Darwin Core Event hierarchy rooted at `TheWilds_2025` and are joined by the SmartWilds collection **[TODO: confirm collection URL]**.

Dataset Details

- **Curated by:** Jenna M. Kline, Anirudh Potlapally, Elizabeth G. Campolongo
- **Homepage:** https://github.com/Imageomics/wilds_multimodal_data
- **Paper:** SmartWilds: A Systems Dataset and Metadata Foundation for Cross-Modal Ecological Sensing (in preparation)
- **Snapshot:** `darwin_core_latest_20261008`, release build 2026-10-22

Campaign	Dates	Sessions	Platform
1	Jul 1, Jul 3	6	Parrot Anafi 4K
2	Aug 11, 18, 25	4 + 1	Parrot Anafi 4K; DJI (Aug 11)
3	Sep 8	4	DJI Air 2

Darwin Core tables. The Event table has 55 rows: 1 `Project`, 15 `SensorDeployment` flight sessions, and 39 `MediaEvent` videos. There are 318,203 Occurrences, each one a detection crop on a sampled frame. `metadata/flights.csv` summarizes each session with its platform, number of videos, and number of Occurrences.

The 15 sessions are the 15 directories that contain video. Three more Aug 25 Parrot directories contain telemetry but no video. Their files are included in `media/`, but they are not sampling Events.

Supported Tasks

- **Aerial detection and classification:** you can re-label the raw video and compare against the released crops.
- **Telemetry-aware vision:** altitude, heading, and gimbal pitch are available per frame where the platform logged them.
- **Workload studies:** detection arrivals are extremely bursty (B ≈ +0.99), so they are a stress case for scheduling and offload.

Dataset Structure

```
smartwilds-drones/
├── data/
│   ├── events.csv          # Project, 15 flight SensorDeployments, 39 MediaEvents
│   ├── occurrences.csv     # 318,203 frame-level detection crops
│   └── detections.csv      # per detection: video path, frame index, box (px)
├── media/                  # media/<campaign>/<flight>/... videos + telemetry
├── media_manifest.csv
├── metadata/
│   ├── flights.csv         # per-session platform, video and occurrence counts
│   ├── platforms.csv       # resolution, telemetry format per platform
│   ├── drone_projection_config.csv
│   └── provenance/         # date, coordinate, media and excluded-video audits
├── analysis/bird_water_association_per_box.csv
└── scripts/
```

Raw video, not frames

The release ships the **raw videos**. It does not ship extracted frames. Each machine detection in `data/detections.csv` gives the video (`video_path`), the native frame number (`frame_index`), and the box in pixels. SAM3 ran on every 15th frame. Any frame can be regenerated:

```
import cv2, pandas as pd
det = pd.read_csv("data/detections.csv")
row = det.iloc[0]
cap = cv2.VideoCapture(row.video_path)
cap.set(cv2.CAP_PROP_POS_FRAMES, int(row.frame_index))
ok, frame = cap.read()
x, y, w, h = map(int, (row.bbox_x, row.bbox_y, row.bbox_w, row.bbox_h))
crop = frame[max(y, 0):y + h, max(x, 0):x + w]
```

The 644 curated goose records come from a separate 20-frame review subset and have no `frame_index`.

Annotation method

- **Method:** automated, with a human-validated subset.
- **Detection:** SAM3 on frames extracted from each video.
- **Classification:** top-1 label from BioCLIP Tree-of-Life, written to the Darwin Core taxon fields. The rank, score, box, and altitude are in `identificationRemarks`.
- **Ground projection:** the ray through the bottom-center of the box is intersected with a flat ground plane, using drone GPS, AGL altitude, heading, gimbal pitch (with a fallback), and approximate FOV. `drone_georef_status` in `occurrenceRemarks` gives the result for each detection:

Status	Rows
`ok`	247,340

Status	Rows
`invalid_altitude`	27,306
`missing_drone_gps`	24,467
`horizon_ray`	18,324
`out_of_range`	122

When the status is not `ok`, the coordinate is the drone's own position, a validated static video-level location (500 m uncertainty), or blank. `dataGeneralizations` says which on every row. The flights were flown for population counts and behavior, not geolocation, so platform position places an animal in the right pasture.

- **Verification:** 317,559 rows are `unverified`. 644 are `validated` Canada goose records from a 20-frame review of `20250703_parrotanafi_5`.

Dataset Creation

Known defects and repairs

- **Aug 11 session has no time of day.** Its `eventDate` was exported as the string `nan` (41,524 Occurrences). The date was restored from the flight identifier. `eventTime` is blank because neither the telemetry nor the video containers keep a trustworthy capture time. Do not use these rows for intra-day or inter-arrival analyses.
- **Impossible coordinates on 7 Events** (`nan`, or longitude ≈ 500). These were replaced with a validated static video location, or left blank.
- **Frame media lost from scratch storage.** Extracted frames were deleted when the scratch workspace was purged. The affected Occurrences now point at the original video and are tagged `associated_media_source=original_video_fallback_missing_scratch_frame`. The release build on 2026-10-22 applied the same fallback to the last 10,943 rows (479 frames). As a result, every drone `associatedMedia` value resolves to a released video.
- **One video was not processed.** `20250703_parrotanafi_4/VID_001` was reviewed and found to have no usable content. It is a `MediaEvent` with zero Occurrences.
- **Three telemetry schemas.** Field names, units (feet vs. meters), and gimbal conventions differ by platform and were mapped per platform.

Personal and sensitive information

3,114 Occurrences are labelled *Homo sapiens*: field staff and visitors seen from altitude. **[TODO(release): decide on blurring/withholding policy for frames with people.]**

Considerations for Using the Data

- **Detections count crops, not animals.** Adjacent frames repeat the same animal, so the counts measure evidence volume, not abundance.
- **Species-rank labels need curation.** The class-rank accuracy reported for the camera-trap subset (0.845) is the better guide to these labels.
- **The DJI airframe for Aug 11 was not recorded.**

Supplemental Material

Extended inventories, the timestamp-repair methods, the standards crosswalk, validation checks, and the full defect inventory are in `SUPPLEMENTAL.md` in the study-level SmartWilds dataset. **[TODO: confirm repo URL]**

Licensing

Data: CC0 1.0 **[TODO(license): confirm across the collection]**. Scripts: MIT.

Citation

```
@misc{smartwilds2025_drones,
  author    = {Kline, Jenna M. and Potlapally, Anirudh and Campolongo, Elizabeth G.},
  title     = {SmartWilds Drones (The Wilds 2025)},
  year      = {2026},
  publisher = {Hugging Face},
  doi       = {TODO},
  url       = {https://huggingface.co/datasets/imageomics/smartwilds-drones}
}
```


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Dataset Card Contact

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Appendix: Hugging Face metadata (YAML front matter)

```yaml
license: cc0-1.0
language:
- en
pretty_name: SmartWilds Drones (The Wilds 2025)
task_categories:
- object-detection
- image-classification
tags:
- biology
- ecology
- biodiversity
- drone
- uav
- aerial-imagery
- darwin-core
- humboldt-eco
- fair2
- smartwilds
- the-wilds
- ohio
size_categories:
- 100K<n<1M
description: >-
  39 drone videos (95 GB) from 15 flight sessions on 6 dates at The Wilds
  (Ohio, USA), July-September 2025, from three platforms (Parrot Anafi 4K, DJI,
  DJI Air 2), with flight telemetry and 318,203 Darwin Core Occurrences
  (frame-level SAM3 + BioCLIP Tree-of-Life detection crops with projected
```

```
  ground positions).

fair2_compliance:
  findable:
    doi: ""  # TODO: assign on publication
    metadata_registry: ["DataCite", "GBIF"]
  accessible:
    open_access: true
    authentication_required: false
  interoperable:
    standards: ["Darwin Core", "Humboldt Eco", "FAIR2"]
  reusable:
    license_clear: true
    provenance_documented: true
  ai_ready:
    machine_readable: true
    structured_annotations: true

darwin_core:
  event_coverage:
    start_date: "2025-07-01"
    end_date: "2025-09-08"
    decimal_latitude: 39.83
    decimal_longitude: -81.74
    coordinate_uncertainty_meters: 5000
    locality: "The Wilds, Cumberland, Muskingum County, Ohio, USA"
  occurrence_info:
    basis_of_record: "MachineObservation"
    sampling_protocol: "Manual drone video survey of pastures and lake margin; SAM3 detection on sampled fram

platform:
  type: "UAV"
  models: ["Parrot Anafi 4K", "DJI (airframe unrecorded)", "DJI Air 2"]
  autonomy_mode: "manual"

sensors:
  - type: "RGB"
    platform: "Parrot Anafi 4K"
    video: "4K MP4"
    telemetry: "Parrot JSON / GUTMA DX"
  - type: "RGB"
    platform: "DJI (airframe unrecorded)"
    video: "1080p MOV"
    telemetry: "CSV, 10 Hz"
  - type: "RGB"
    platform: "DJI Air 2"
    video: "4K MP4"
    telemetry: "Airdata CSV"

mission:
  telemetry_available: true

configs:
- config_name: occurrences
  data_files: data/occurrences.csv
- config_name: events
```

```
  data_files: data/events.csv
- config_name: flights
  data_files: metadata/flights.csv
```